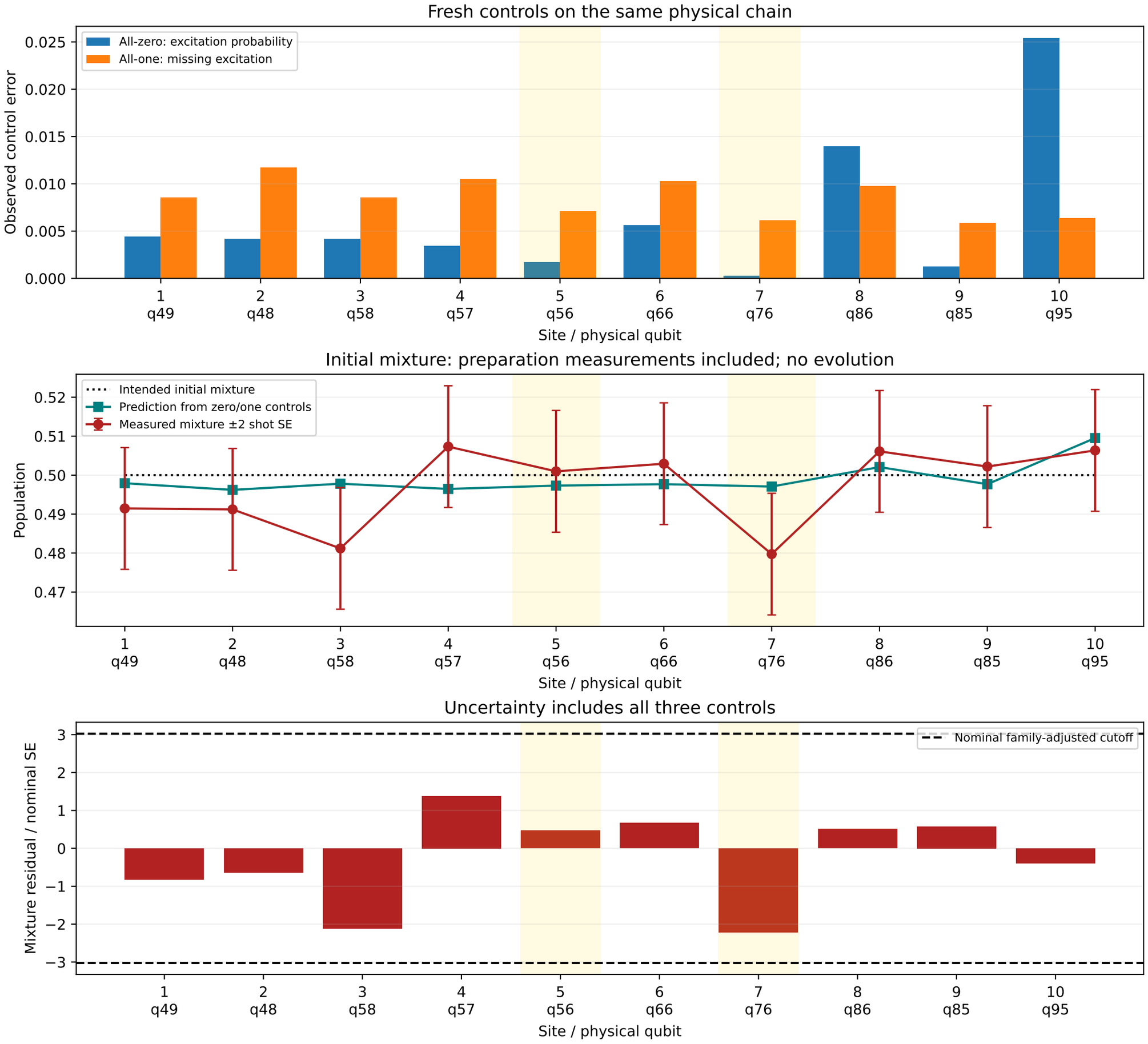
Fresh controls on the same physical chain
All-zero: excitation probability
All-one: missing excitation
Observed control error
0.025
0.020
0.015
0.010
0.005
0.000
1 q49
2 q48
3 q58
4 q57
5 q56
6 q66
7 q76
8 q86
9 q85
10 q95
Site / physical qubit
Initial mixture: preparation measurements included; no evolution
Intended initial mixture
Prediction from zero/one controls
Measured mixture ±2 shot SE
Population
0.52
0.51
0.50
0.49
0.48
0.47
1 q49
2 q48
3 q58
4 q57
5 q56
6 q66
7 q76
8 q86
9 q85
10 q95
Site / physical qubit
Uncertainty includes all three controls
Nominal family-adjusted cutoff
Mixture residual / nominal SE
3
2
1
0
−1
−2
−3
1 q49
2 q48
3 q58
4 q57
5 q56
6 q66
7 q76
8 q86
9 q85
10 q95
Site / physical qubit